AMERICAN INTERNATIONAL UNIVERSITY -BANGLADESH
Department of Computer Science
FACULTY OF SCIENCE and TECHNOLOGY
Program: CSE
Course Name: PROGRAMMING IN PYTHON
SECTION: C

Semester: SPRING 2025-2026	**Date of Submission:** 28.03.26

MID PROJECT REPORT

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SUBMITTED TO
MD. TANZEEM RAHAT
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SUBMITTED BY:

ID	NAME
22-49494-3	ARIA RAHMAN
22-48884-3	MD. SAJJAD MAHMUD NAYON

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Faculty comments: ……
Signature: …… **Date:** ……

SPACE SHOOTER GAME

Project overview:

This project is under game track. In space shooter game, the players can control the spaceship and destroy the alien waves. The player earns 10 points per kill. The invaders need to be shot at least 3 times and if the game is over, the player can restart the game and beat the high scores. This project also covers OOP design, event-driven game loops, collision detection, a full menu system, and CSV-based persistent player data.

Setup & Run Steps:

- Python 3.10+ installed and added to PATH
- All game assets present in the images/ folder
- requirements.txt included — generated via pip freeze > requirements.txt
- **Install dependencies:** pip install -r requirements.txt
- **Run the game:** python main.py

Feature List:

Feature	Details
Main Screen	2 buttons: Start Game, Leaderboard
Player Name Entry	Input box, validation, duplicate check, best score shown
CSV File	Players.csv has name + score
How to play	Shown as instructions
Countdown Timer	Counts 3 to 1 to GET READY before each play
Alien Grid	Alien moves left to right & has 5 columns with 4 rows
Score System	10 points for per invaders to be killed & saved to CSV
Collision Detection	Spaceship must shoot 3 types to destroy the invader
Heart power-up	Randomly drops and collecting them can restore 1 health
Sheild	Earned on kill and blocks invader's bullets for 3 seconds
Explosion	3 sizes animated sprites on kills and death for spaceship
Quit Button	Red, top-right, saves score, returns to menu
Restart Button	Yellow, end screen, saves score, new game
High Scores	Top 5 players from CSV file and to return use ESC

Screenshots:

Figure 1: main window

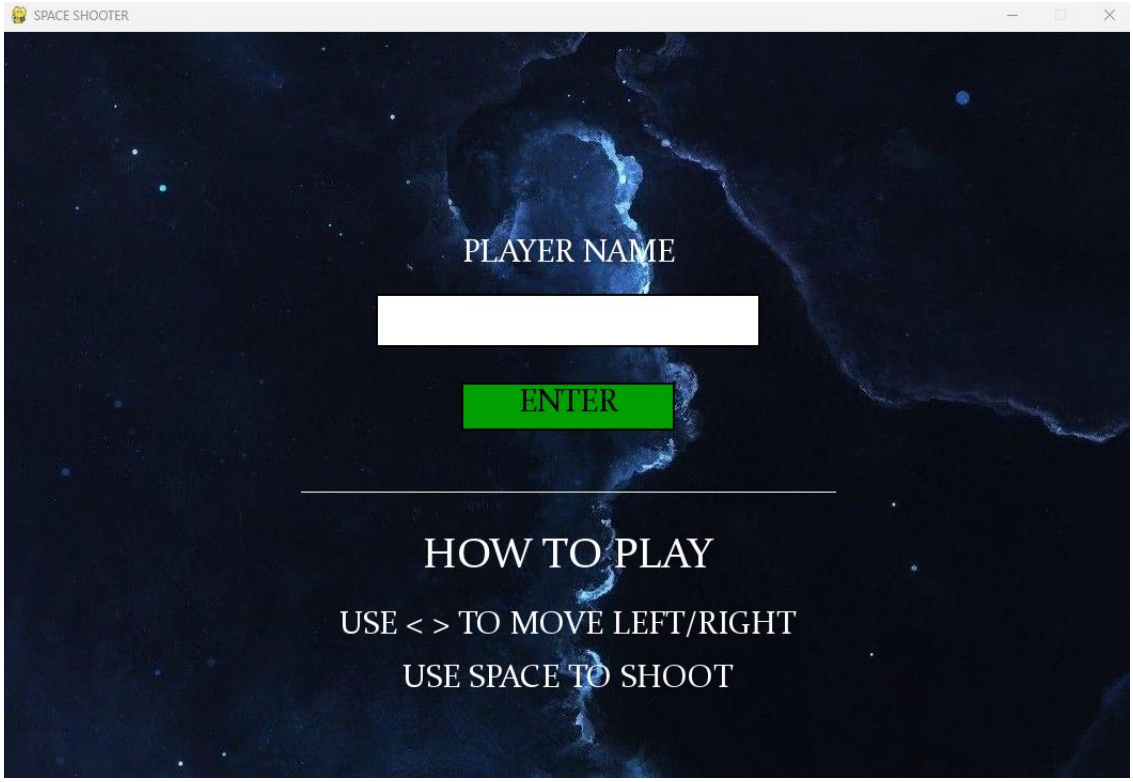

Figure 2: player entry & instruction

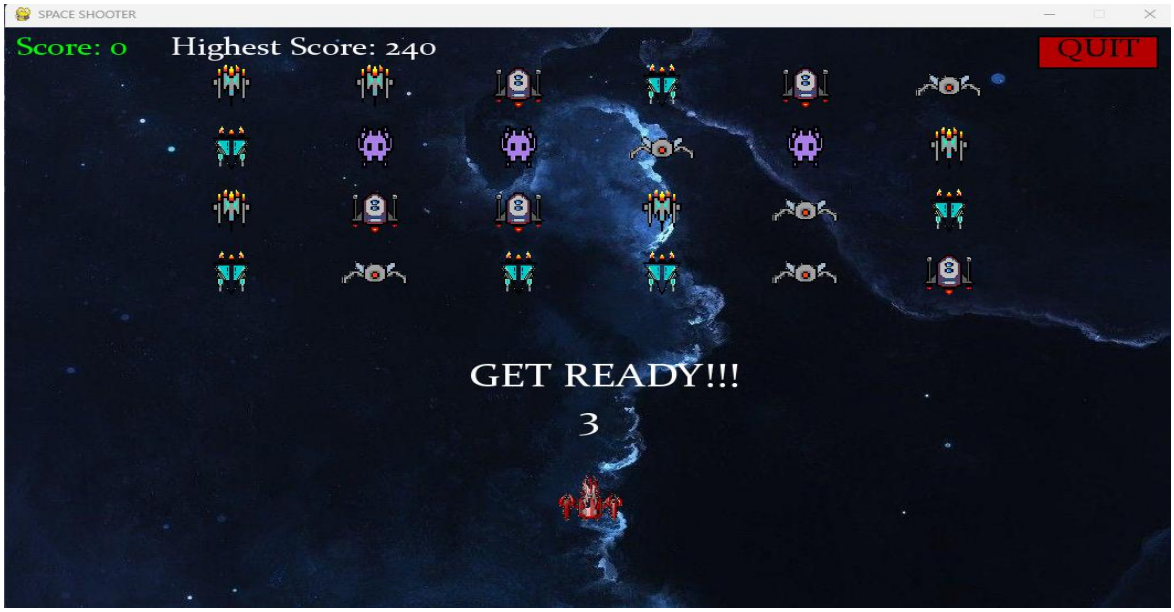

Figure 3: in game window, countdown before the game starts

Figure 4: in game window, live score on top left & quit, shield and gaining lives

Figure 4: game over, final score, restart

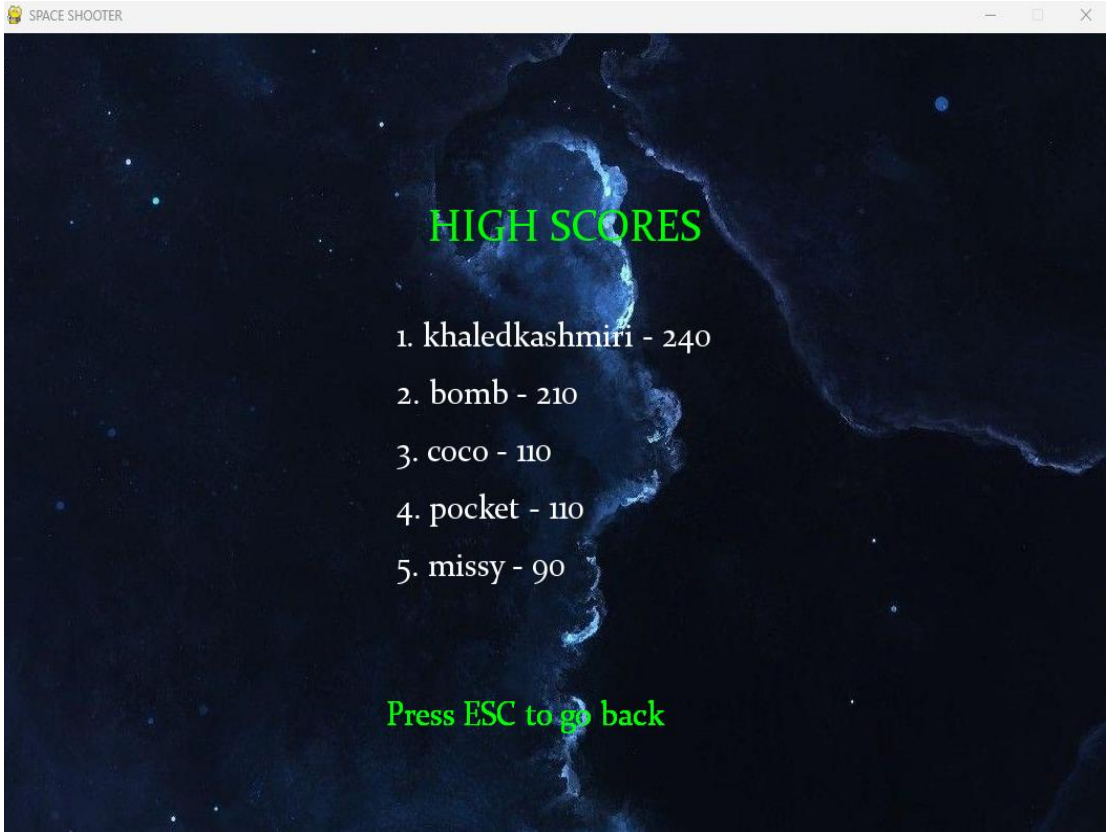

Figure 5: top 5 players with high scores